Dealing with Heteroskedasticity and Dependence

Applied Regression in R

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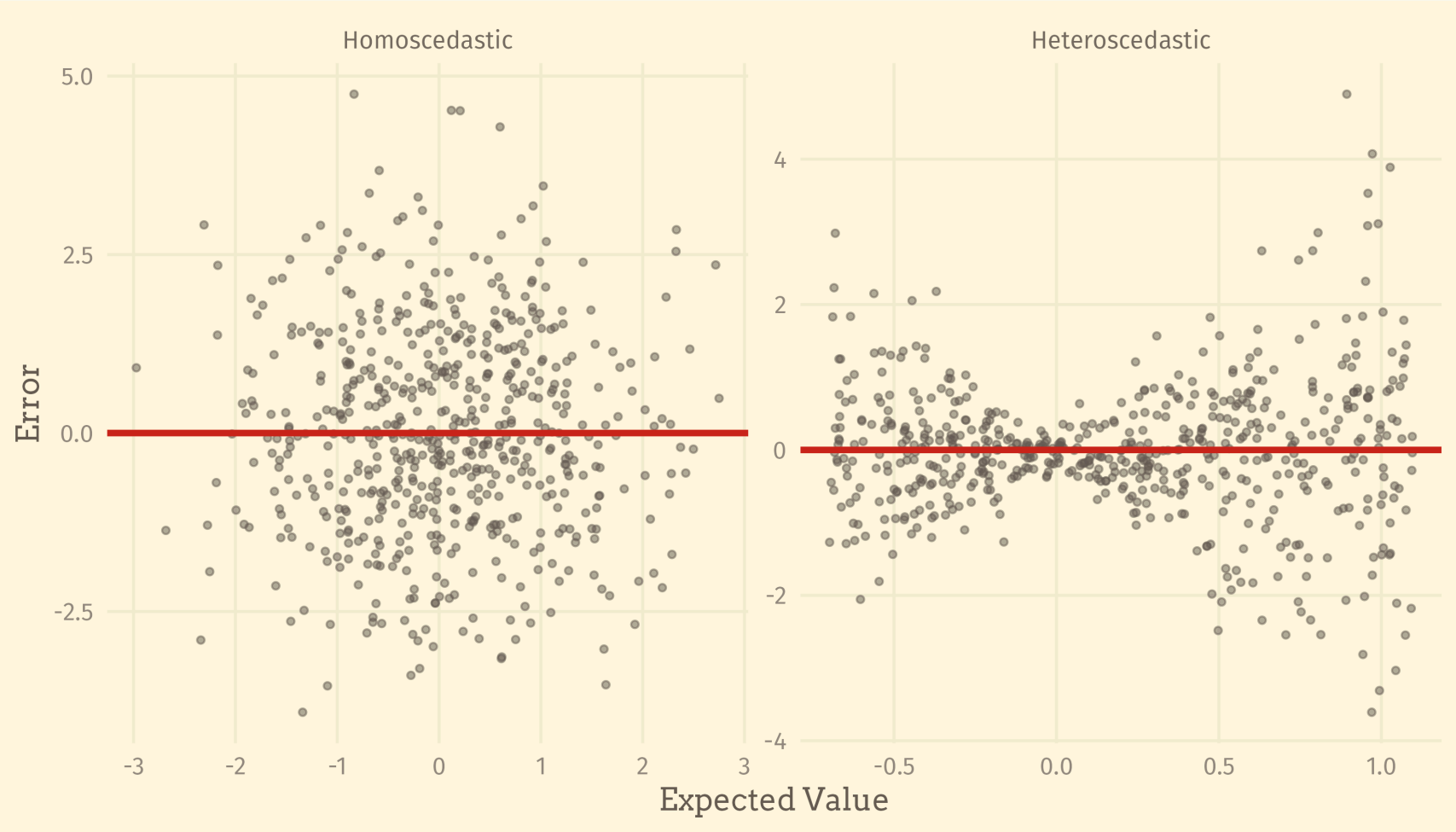
Today's Goals

What are:

1. Robust standard errors?
2. Clustered standard errors?
3. Bootstrap standard errors?

Homoscedastic vs. Heteroscedastic Residuals



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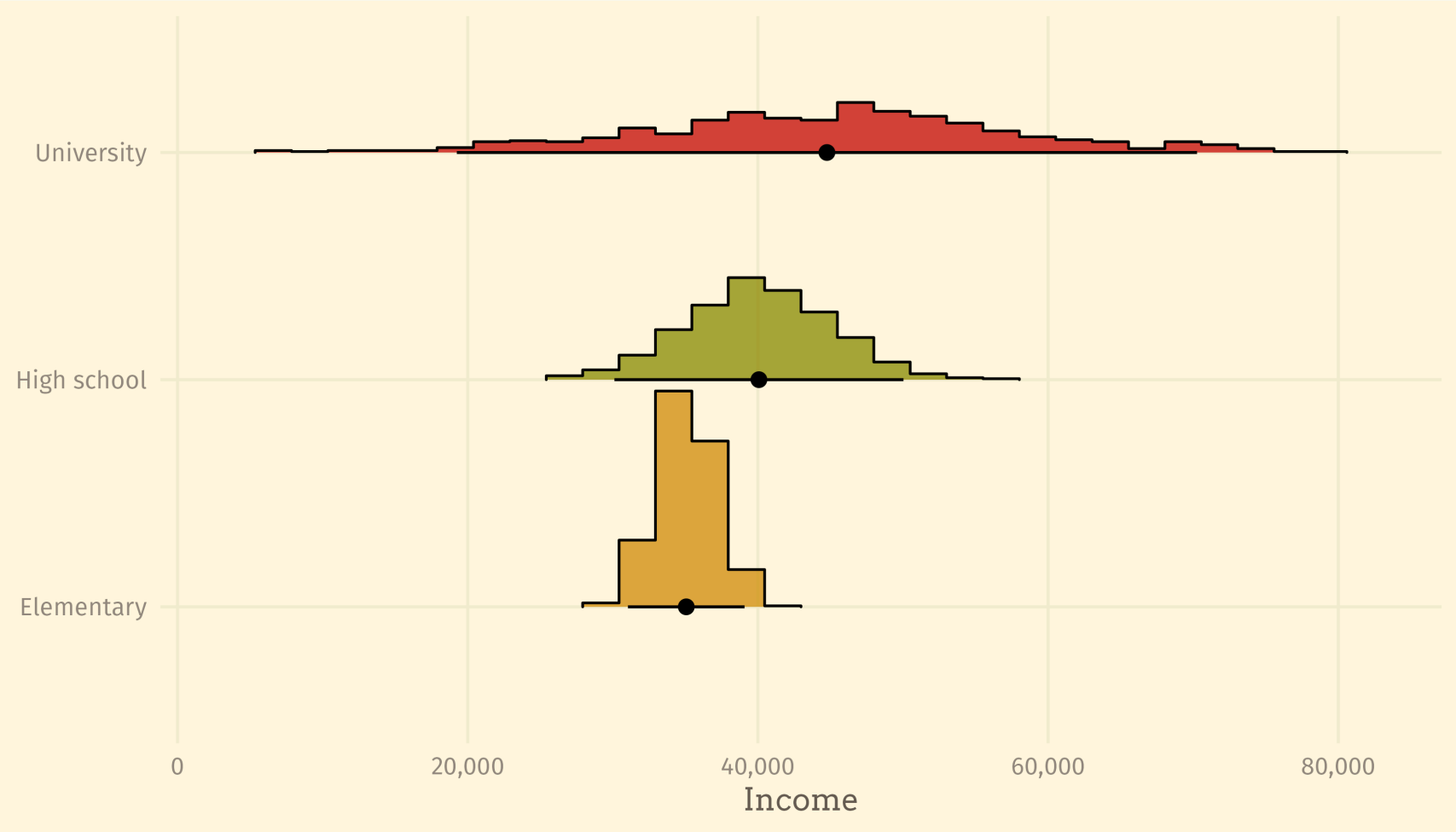

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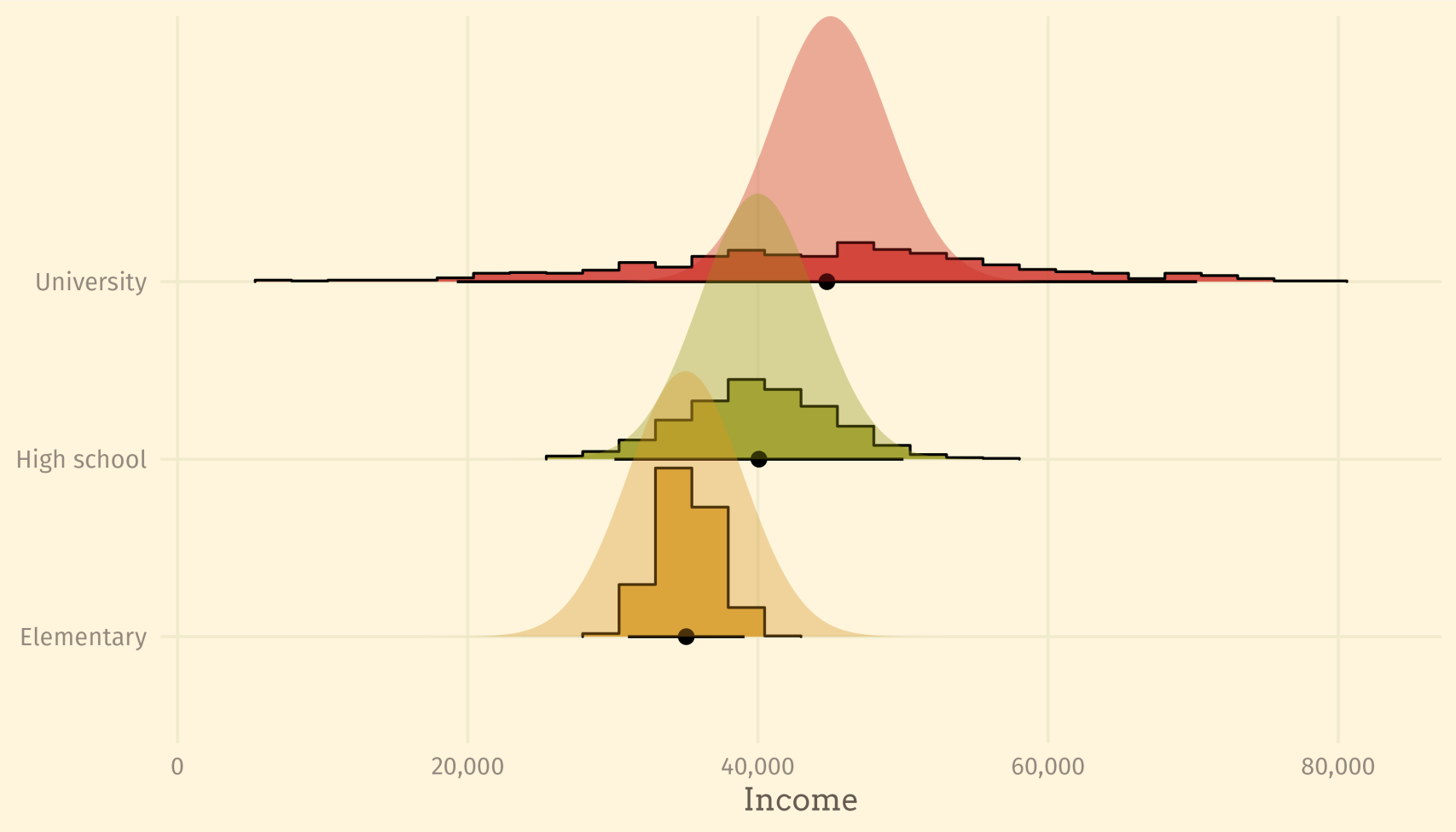
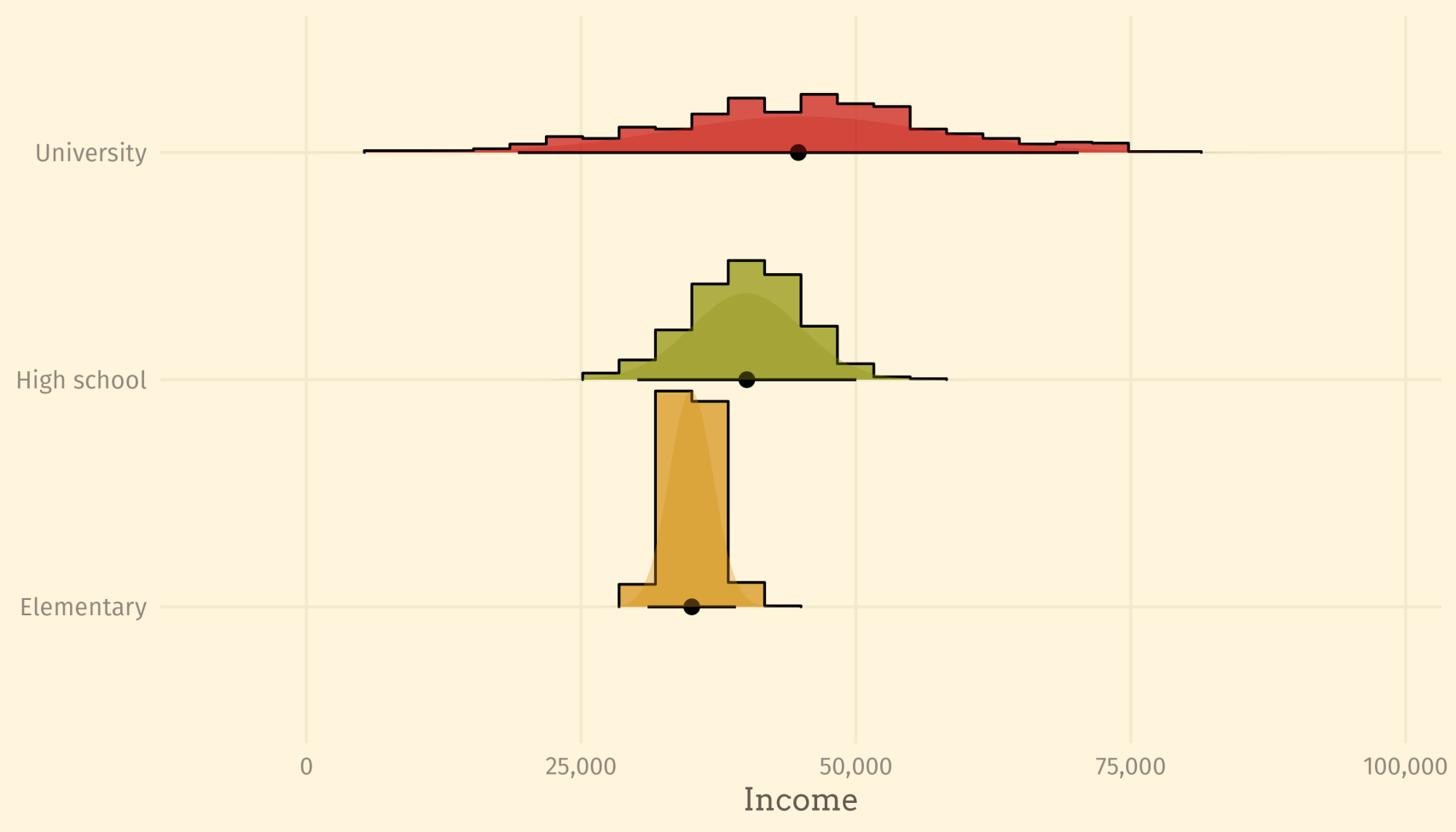


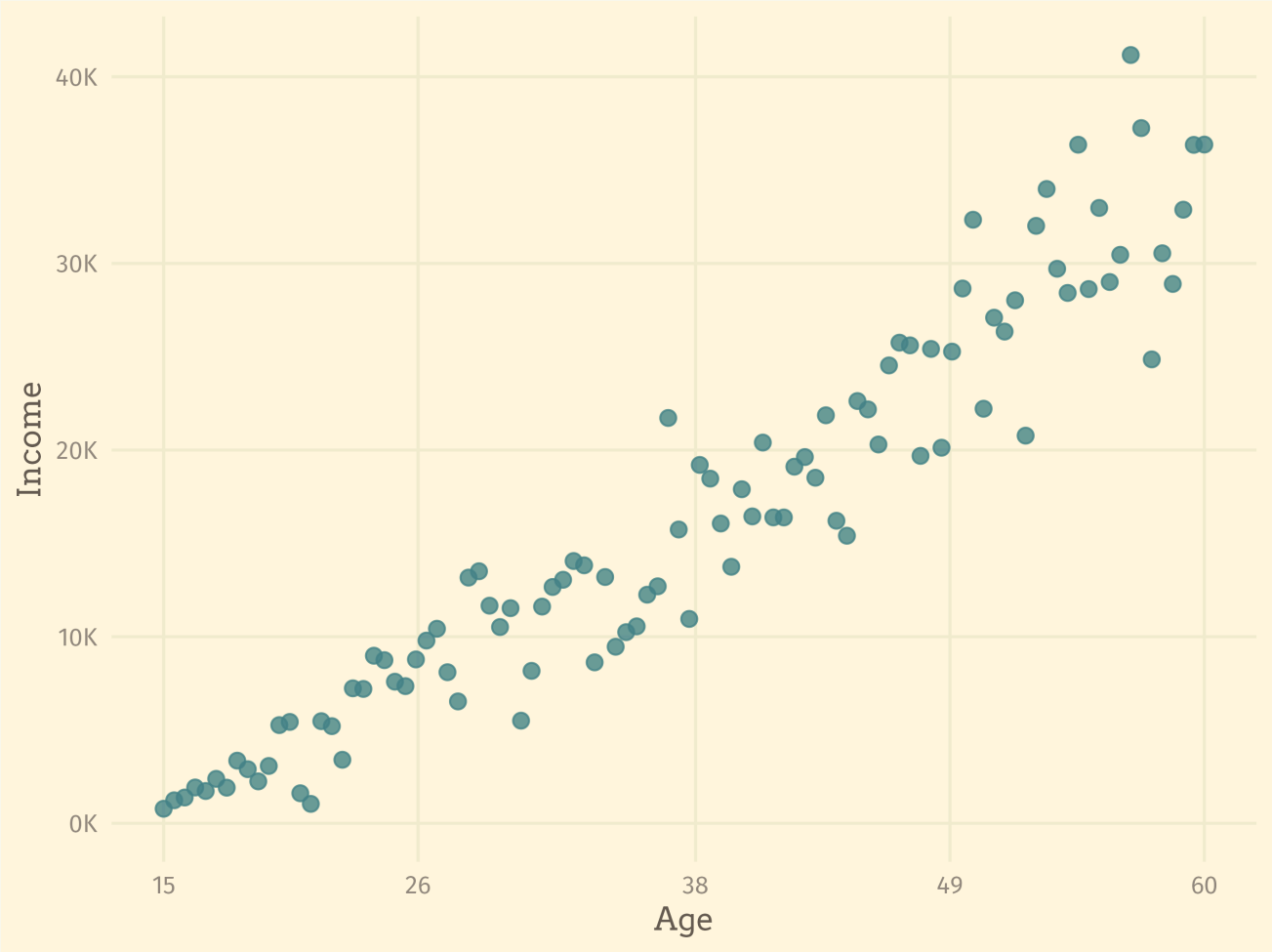
Solution?

Estimate variances separately

What is Homoscedasticity?



Heteroscedasticity Also Arises in Continuous Predictors



What is Homoscedasticity?

When residual variance is (roughly) equal, assuming homoscedasticity makes model estimation **faster and more simple**.

However, assuming homoscedasticity will **bias standard errors**, if the assumption is not actually true.

What is Homoscedasticity?

Standard errors that don't assume homoscedasticity are called **robust standard errors**.

Many different types, with funny names — HC0, HC1, HC2, HC3, ...

You can use pretty much any you want, except for HC0.

Questions?

The Horrors of Variance--Covariance Matrix

Variance-Covariance Matrix

The main part of regression:

$$y = \beta_0 + \beta_1 \cdot x + \varepsilon$$

Variance-Covariance Matrix

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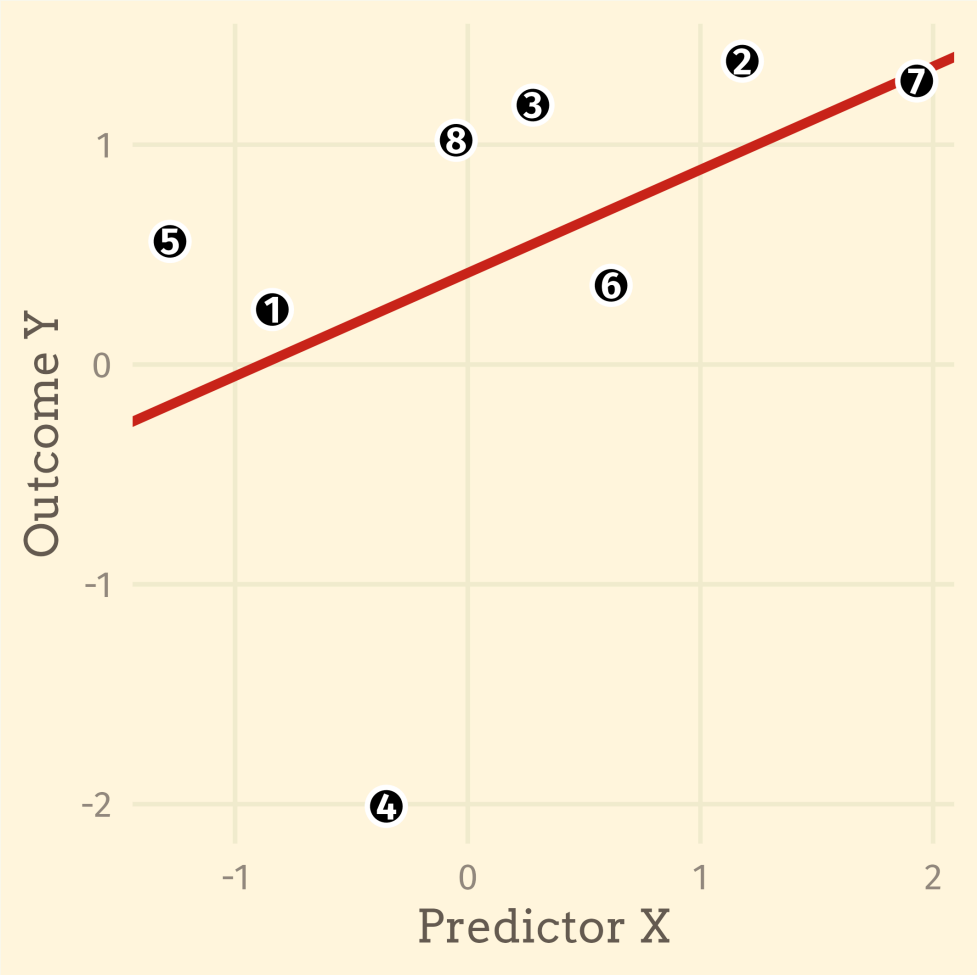
$$y = \beta_0 + \beta_1 \cdot x + \varepsilon$$

Information about standard errors etc., stored in Variance-Covariance matrix.

Residual Standard Errors

ID	X	Y
1	-0.84	0.25
2	1.18	1.38
3	0.28	1.18
4	-0.35	-2.01
5	-1.28	0.56
6	0.616	0.36
7	1.93	1.29
8	-0.05	1.02



Residual Variance-Covariance Matrix

For every observed value of the dependent variable, we store estimated residual standard deviation in a **variance-covariance matrix**.

1.13	0	0	0	0	0	0	0
0	1.13	0	0	0	0	0	0
0	0	1.13	0	0	0	0	0
0	0	0	1.13	0	0	0	0
0	0	0	0	1.13	0	0	0
0	0	0	0	0	1.13	0	0
0	0	0	0	0	0	1.13	0
0	0	0	0	0	0	0	1.13

Residual Variance-Covariance Matrix

Values on the diagonal are estimated **standard deviations of residuals**.

(The are all the same because we assume it.)

1.13	0	0	0	0	0	0	0
0	1.13	0	0	0	0	0	0
0	0	1.13	0	0	0	0	0
0	0	0	1.13	0	0	0	0
0	0	0	0	1.13	0	0	0
0	0	0	0	0	1.13	0	0
0	0	0	0	0	0	1.13	0
0	0	0	0	0	0	0	1.13

Residual Variance-Covariance Matrix

Values off the diagonal are estimated **standard covariances of residuals**.

(They are all 0, because we assume independence.)

1.13	0	0	0	0	0	0	0
0	1.13	0	0	0	0	0	0
0	0	1.13	0	0	0	0	0
0	0	0	1.13	0	0	0	0
0	0	0	0	1.13	0	0	0
0	0	0	0	0	1.13	0	0
0	0	0	0	0	0	1.13	0
0	0	0	0	0	0	0	1.13

Residual Variance-Covariance Matrix

Classical linear regression assumes that all residual variances are the same. This is the assumption of **homoskedasticity**.

It also assumes all residual covariances are zero. This is the assumption of **independence of errors**.

1.13	0	0	0	0	0	0	0
0	1.13	0	0	0	0	0	0
0	0	1.13	0	0	0	0	0
0	0	0	1.13	0	0	0	0
0	0	0	0	1.13	0	0	0
0	0	0	0	0	1.13	0	0
0	0	0	0	0	0	1.13	0
0	0	0	0	0	0	0	1.13

Robust Standard Errors

We can estimate residual variance **separately** for each value of the dependent variable.

This produces robust standard errors.

0.09	0	0	0	0	0	0	0
0	0.27	0	0	0	0	0	0
0	0	0.66	0	0	0	0	0
0	0	0	8.5	0	0	0	0
0	0	0	0	0.92	0	0	0
0	0	0	0	0	0.19	0	0
0	0	0	0	0	0	0.001	0
0	0	0	0	0	0	0	0.65

Clustered Robust Errors

We have dealt with heteroskedasticity, but what if the errors aren't independent?

We can estimate the **covariance between them** as well.

The result is known as **clustered robust standard errors**.

0.09	0.23	.11	0.4	0	0	0	0
0.23	0.27	0.5	0.29	0	0	0	0
.11	0.5	0.66	0.09	0	0	0	0
0.4	0.29	0.09	8.5	0	0	0	0
0	0	0	0	0.92	0.07	0.18	0.41
0	0	0	0	0.07	0.19	0.91	0.22
0	0	0	0	0.18	0.91	0.001	0.03
0	0	0	0	0.41	0.22	0.03	0.65

Questions?

InteRmezzo!

Why Not Use (Clustered) Robust SE All the Time?

Why Not Use (Clustered) Robust SE All the Time?

Actually pretty common in economics.

If using linear regression with big sample size, not a bad strategy.

Two potential problems:

- **Biased for small samples.**
- **Less efficient** than classical errors — more parameters to estimate.

Why Not Use (Clustered) Robust SE All the Time?

Using clustered Errors by default is more problematic. Two problems:

- Clustered standard errors **biased for small number of clusters.**
- Clustering for the incorrect variable **just lowers precision** with no bias correction — you've just inflated your SE for no reason.

The More Clusters, The Less Parameters To Estimate

Four Clusters

0.09	0.23	0	0	0	0	0	0
0.23	0.27	0	0	0	0	0	0
0	0	0.66	0.09	0	0	0	0
0	0	0.09	8.5	0	0	0	0
0	0	0	0	0.92	0.07	0	0
0	0	0	0	0.07	0.19	0	0
0	0	0	0	0	0	0.001	0.03
0	0	0	0	0	0	0.03	0.65

Two Clusters

0.09	0.23	.11	0.4	0	0	0	0
0.23	0.27	0.5	0.29	0	0	0	0
.11	0.5	0.66	0.09	0	0	0	0
0.4	0.29	0.09	8.5	0	0	0	0
0	0	0	0	0.92	0.07	0.18	0.41
0	0	0	0	0.07	0.19	0.91	0.22
0	0	0	0	0.18	0.91	0.001	0.03
0	0	0	0	0.41	0.22	0.03	0.65

Bootstrapping

Bootstrapping

To compute standard errors, we need to know what the **sampling distribution** of our parameters looks like.

We can either:

- Approximate it using analytical formulas,
- Or **estimate it directly from the data.**

Bootstrapping

https://seeing-theory.brown.edu/frequentist-inference/index.html#section3

Bootstrapping

If our sample is representative (looks like the population) … we can **treat it as the population!**

Repeatedly draw „new“ samples from our data. Differences between samples give an estimate of sampling error.

Bootstrapping allows computing standard errors **without knowing any formulas** — which is a pretty big deal.

Limits of Bootstrap

Strongly assumes that data are representative: the distribution of the sample is the same as in the population. Not generally true for small samples.

Takes a long time to compute for big samples and complex models.

Best used for medium-sized samples and when nothing else works.

Questions?

InteRmezzo!